Model	Version	GSM8K	MATH-L3&L4	MMLU-high	MathOAI
Llama3.1-8B-it	Original prompt	60.33	**21.46**	24.81	30.20
	SAE-based-steering (ours)	**62.67**	**21.46**	**27.04**	**32.80**
DeepSeek-Llama3-8B	Original prompt	82.67	78.11	72.96	78.60
	SAE-based-steering (ours)	**85.67**	**79.83**	**76.30**	**83.40**
Gemma-2-9b-it	Original prompt	88.00	**51.07**	51.11	60.20
	SAE-based-steering (ours)	**90.67**	50.64	**52.22**	**62.60**

Table 1: Overall accuracy (%) of steering.

task-irrelevant features are very close between both processes shown in Figure 2. We therefore propose to compute the *absolute difference* between activations, effectively suppressing noisy features while preserving relevant features. Specifically, the final CoT feature activation for question q can be described as $\alpha_t = \|\alpha_t^x - \alpha_t^y\|$.

To better explain the motivation of our method. We derived the top-10 feature activations of the Verbal Reasoning Process and the Symbolic Reasoning Process by several representative examples. The result is shown as Figure 2. For each subfigure, the feature indexs are categorized into three groups, the left group represents features predominantly associated with verbal featurs ($\alpha_t^x \gg \alpha_t^y$), features in the middle group indicates balanced features between both processes, the right group represents features predominantly associated with symbolic features ($\alpha_t^x \ll \alpha_t^y$). We can find that for `DeepSeek-Llama3-8B`, features in the middle group, such as 19555 and 14602 mean "punctuation marks, specifically commas" which is unrelated to reasoning; features index 462 in the left group present big difference between two processes, it means "relationship statements involving variables"; features in the right group like 15593 means "numeric expressions and mathematical components". For `Gemma-2-9b-it`, the phenomenon is almost the same as the former: feature 5012 and feature 14451 in the middle group are unrelated to reasoning. These extracted results are the same as the assumption we made at first, which means subtraction can better erase noisy features while integrating useful information.

In practice, we randomly sample $N = 100$ samples:$\{q_p\}_{p=1}^N$ from the MATHOAI dataset[3] to generate the Verbal Reasoning Process $\{x_p\}_{p=1}^N$ and the Symbolic Reasoning Process $\{y_p\}_{p=1}^N$. The reason for selecting this dataset is that we want to ensure that the difficulty of the problem is not too complicated to avoid generating long CoTs, and not too simple to avoid generating indistinguishable CoTs for both processes. The final feature activation $\boldsymbol{\alpha}_t$ derived from $\{q_k\}_{l=1}^N$ can be described as follows:

$$\alpha_t = \frac{1}{N}\sum_{p=1}^{N} \|\alpha_t^{x_p} - \alpha_t^{y_p}\| \tag{2}$$

2.4 Steering

After obtaining $\{\alpha_t\}_{t=1}^{d_{\text{SAE}}}$ from the last step, we sort it in descending order to get the top-k important SAE features. We leverage a certain feature among them for steering, which modulates the residual activation of LLMs to control their behaviors. For a given feature indexed by t, inspired by Templeton et al. (2024); Farrell et al. (2024); Galichin et al. (2025), we proposed to use the activation value of $\boldsymbol{\alpha}$ to modify the residual activation $\boldsymbol{r}^h$ as follows:

$$\hat{\boldsymbol{r}}_i^h = \boldsymbol{r}_i^h + \lambda \alpha_t \boldsymbol{z_t}, i \in \mathcal{H} \tag{3}$$

The modification is made in the layer l according to the SAE version we choose, λ is the hyperparameters of strength, $\boldsymbol{z}_t = \boldsymbol{W}_{\text{dec},t} = \boldsymbol{W}_{\text{dec}}[t, :]$.

3 Experiments of SAE-based Steering

3.1 Experimental Settings

Datasets. We evaluate the effect of our steering method on four mathematical benchmarks, including GSM8K (Cobbe et al., 2021), MMLU-high school mathematics (MMLU-high) (Hendrycks et al., 2021), MATH-500 (Lightman et al., 2023)[4] and MATHOAI[5]. For the MATH-500 dataset, we

[3] https://huggingface.co/\datasets/heya5/math_oai

[4] https://huggingface.co/datasets/HuggingFaceH4/MATH-500

[5] https://huggingface.co/datasets/heya5/math_oai

Model	Version	GSM8K	MATH-L3&L4	MMLU-high	MathOAI
DeepSeek-Llama3-8B	Original prompt	82.67	78.11	72.96	78.60
	BoostStep (Zhang et al., 2025)	79.33	77.25	71.11	78.80
	Mathneuro (Christ et al., 2024)	84.67	78.54	68.89	76.00
	SAE-free-steering (Ours)	**85.67**	**82.83**	**78.15**	**83.00**
DeepSeek-qwen-1.5B	Original prompt	70.00	65.24	65.56	74.20
	BoostStep (Zhang et al., 2025)	67.33	66.09	63.70	73.40
	Mathneuro (Christ et al., 2024)	72.66	66.09	63.33	73.00
	SAE-free-steering (Ours)	**74.00**	**69.10**	**66.67**	**77.00**

Table 2: Performance comparison (accuracy %) between original LLMs and SAE-Free steering versions.

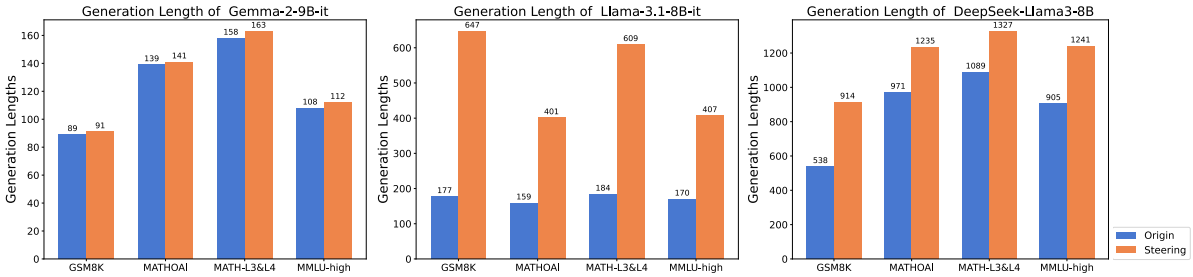

Figure 3: The generation length variation of the questions with the top 25% shortest answers generated by the original LLM.

select MATH-level3 and MATH-level4 for testing. For the GSM8K dataset, to find the best parameters and features more efficiently, we conduct our experiment on a subset that contains 300 randomly selected samples instead of on the entire dataset. The baseline results are detailedly shown in appendix Table 3. It means the results of subsets are roughly the same as the results on the entire dataset, thus ensuring that the randomly selected samples can be used to explore the effect of the steering experiment.

LLM Models and SAEs Because we want to evaluate LLMs that have been proven to have excellent thinking ability, but at the same time, our method is based on pre-trained SAE, so we choose the `DeepSeek-Llama3-8B` (Guo et al., 2025) and its base version `Llama3.1-8B-it` (Grattafiori et al., 2024). Besides, we also choose `Gemma-2-9b-it` (Team et al., 2024) as evaluation LLM. For SAEs, we leverage `llama_scope_r1_distill(l15r_800m_openr1_math)`[6] to extract features from `DeepSeek--Llama3-8B` and `Llama3.1-8B-it`, `gemma-scope-9b-it-res(layer_9/width_16k/canonical)`[7] to extract features from `Gemma-2-9b-it`.

Implementation Details. For the generation setting, we employed a repetition penalty of 1.2 while setting distinct maximum sequence lengths for each dataset: 3000 tokens for GSM8K, 2500 for MATH-500, 3000 for MMLU-High, and 3500 for MATHOAI. For evaluation, we leverage LLM as a judge (Phan et al., 2025) to evaluate the accuracy of predictions. We utilize the GPT4.1 API as the judge. More details can be found in Appendix C.

3.2 Experimental Results Analysis

Comparing with Baselines. Table 1 reports the overall accuracy results of four mathematical reasoning datasets. In addition to MATH-L3&L4, there is a consistent improvement in the other three datasets. Especially for `DeepSeek-Llama3-8B`, steering improves the accuracy by 3.34% and 4.80% in MMLU-high and MathOAI. There is no significant improvement on MATHL3&L4 for three LLMs, only `DeepSeek-Llama3-8B` achieves 1.72% improvement, which means that SAE steering may play a better role in the reasoning model compared to the instruction model. Through further

[6] https://huggingface.co/fnlp/Llama-Scope-R1-Distill

[7] https://huggingface.co/google/gemma-scope-9b-it-res

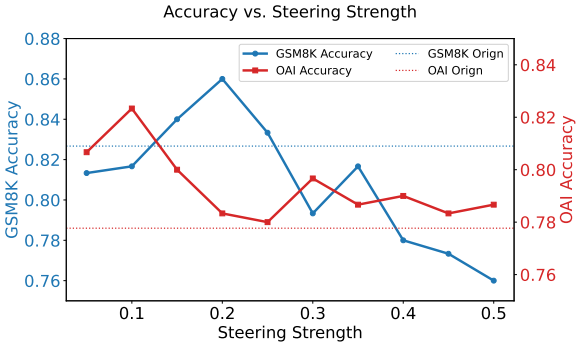

Figure 4: Relationship between strength λ and accuracy in GSM8K and MATHOAI.

manual reviewing, we found that for non-reasoning LLM `Llama3.1-8B-it`, too long generation length will lead the LLM to generate some irrelevant information, steering may improve this phenomenon by generating some thinking process related to the steering feature.

Steering Effects on Reasoning Length. We also compared the length variation of the questions with the top 25% shortest answers generated by the original LLM before and after steering. The result is shown in Figure 3. Our experimental results indicate a steady enhancement in reasoning length for all evaluated datasets, with `Llama3.1-8B-it` exhibiting a significant increase. After observation, we found that the answers after steering will increase the number of reasoning paths and the depth of reasoning.

Relationship between Strength and Accuracy. Taking `DeepSeek-Llama3-8B` as an example, we plotted the relationship between steering strength and accuracy. From Figure 4 we can see that a higher strength does not always improve the accuracy of the model's response. There is an optimal strength, and exceeding it will cause the model's performance to decline. Manually checking the cases found that when the strength is too large, the model will generate many meaningless answers. This may be because excessively large strength will destroy the internal representation of the model.

4 SAE-Free Steering

4.1 Motivation

Feature-based steering has been proven to be useful in various fields of natural language processing, such as machine unlearning (Muhamed et al., 2025; Farrell et al., 2024), improving truthfulness (Marks and Tegmark, 2023), and interpretability (He et al., 2025; Kharlapenko et al.). Although there are some trained SAEs released to help interpret hidden activations of LLMs, such as Gemma Scope (Lieberum et al., 2024) and Llama Scope (He et al., 2024), there are still large amounts of LLMs that do not have matching SAEs to interpret them, such as the Qwen series. However, training an SAE is time-consuming and requires a lot of cost, so it is obviously unlikely to train an SAE for each layer of each LLM.

Based on the above reason, our method is hard to generalize to the scenarios in which there is no corresponding SAE. To solve this problem, we proposed an algorithm to generalize such an SAE-based steering method to LLMs without the corresponding SAE. Intuitively, we will start by assuming the availability of an SAE for a given LLM, and then derive the calculation to see if the SAE components can be approximated by others.

4.2 Framework

Feature activation for virtual SAE. SAE is trained to minimize the reconstruction loss for the hidden activation $\boldsymbol{h}$ of the layer l. Therefore, $\boldsymbol{h}$ can be decomposed as the weighted sum of feature vectors$\{\boldsymbol{z}_t\}_{t=1}^{d_{SAE}}$ and error term $\boldsymbol{\epsilon}$, and α_t^h denotes the activation value of feature t for $\boldsymbol{h}$ at layer l.

$$\boldsymbol{h} = \sum_{t=1}^{d_{\text{SAE}}} \alpha_t^h \boldsymbol{z}_t + \boldsymbol{b} + \boldsymbol{\epsilon} \tag{4}$$

Now we want to steer the model even though there is no corresponding SAE model. In our method, there are two types of CoT: Verbal Process x contains m tokens and Symbolic Process y contains n tokens. We employ the mean representation in the layer l to represent x and y: $\boldsymbol{x} = \frac{1}{m}\sum_{i=1}^{m} \boldsymbol{r}_i^x$,$\boldsymbol{y} = \frac{1}{n}\sum_{j=1}^{n} \boldsymbol{r}_j^y$. Therefore, the l layer's residual activation of x and y can be written as follows according to equation (4) :

$$\begin{aligned}\boldsymbol{x} &= \frac{1}{m}\sum_{i=1}^{m} \boldsymbol{r}_i^{\boldsymbol{x}} = \frac{1}{m}\sum_{i=1}^{m}(\sum_{t=1}^{d_{\text{SAE}}} \alpha_t^{\boldsymbol{r}_i^{\boldsymbol{x}}} \boldsymbol{z}_t + \boldsymbol{b} + \boldsymbol{\epsilon}) \\ &= \sum_{t=1}^{d_{\text{SAE}}} \alpha_t^{\boldsymbol{x}} \boldsymbol{z}_t + \boldsymbol{b} + \frac{1}{m}\sum^{m} \boldsymbol{\epsilon} \end{aligned} \tag{5a}$$

$$\begin{aligned}\boldsymbol{y} &= \frac{1}{n}\sum_{j=1}^{n} \boldsymbol{r}_j^{\boldsymbol{y}} = \frac{1}{n}\sum_{j=1}^{n}(\sum_{t=1}^{d_{\text{SAE}}} \alpha_t^{\boldsymbol{r}_j^{\boldsymbol{y}}} \boldsymbol{z}_t + \boldsymbol{b} + \boldsymbol{\epsilon}) \\ &= \sum_{t=1}^{d_{\text{SAE}}} \alpha_t^{\boldsymbol{y}} \boldsymbol{z}_t + \boldsymbol{b} + \frac{1}{n}\sum^{n} \boldsymbol{\epsilon} \end{aligned} \tag{5b}$$